



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



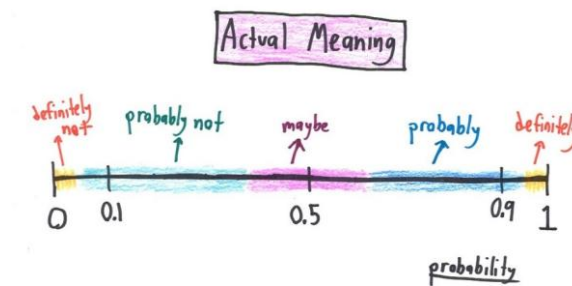


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 9 - Lecture No 17 – March 09, 2026



Announcements!

1. Office hours (Ramadan) – **Tuesday & Thursday 11:30 to 12:30 PM**
2. Quiz No 05 – March 18, 2026
 1. Quizzes – rule for N-1 would be instituted after grading of Quiz No 5 (please note your own standing in the course)
3. Assignment 2 – due March 15, 2026
4. TA's EHSAS Hours

EHSAS TUTORS' RAMADAN SCHEDULE – Spring 2026							
Course	Tutor	Email	Monday	Tuesday	Wednesday	Thursday	Friday
Intro to Probability & Statistics	Rayyan Ali Shah	rs08770@st.habib.edu.pk	2:00-3:00		2:00-3:00		12:40-1:40
	Hiba Aiman Ali	ha09269@st.habib.edu.pk	12:45-2:30		12:45-2:30		11:00-12:00
	Zain Naqi	zn09224@st.habib.edu.pk		9:00 - 10:30	11:45 - 12:45	9:00 - 10:30	

5. Advice**

*ID	Name	Q1 [10 pts] CLO 1	Q2 [15 pts] CLO 3	Q3 [15 pts] CLO 2	Q4 [20 pts] CLO 4	Total [60 pts]	Percentage [100]
	Average	8.8	13.5	8.9	14.3	45.6	76.0
	Percentage	88.4	90.3	59.5	71.6	76.0	

Agenda for today

1. Unit 4 – Discrete Random Variable

Total probability and expectation theorems

- Partition of **sample space** in events, $A_1, A_2, \dots, A_n$
 - ❖ $p_X(x) = P(A_1) p_{X|A_1}(x) + \dots + P(A_n) p_{X|A_n}(x)$
 - ❖ $E[X] = P(A_1) E[X | A_1] + \dots + P(A_n) E[X | A_n]$

$$Y = \{y_1, y_2, \dots, y_n\} \quad A_i = \{Y = y_i\}$$

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y)$$

Total probability theorem

$$E[X] = \sum_y p_Y(y) E[X | Y = y]$$

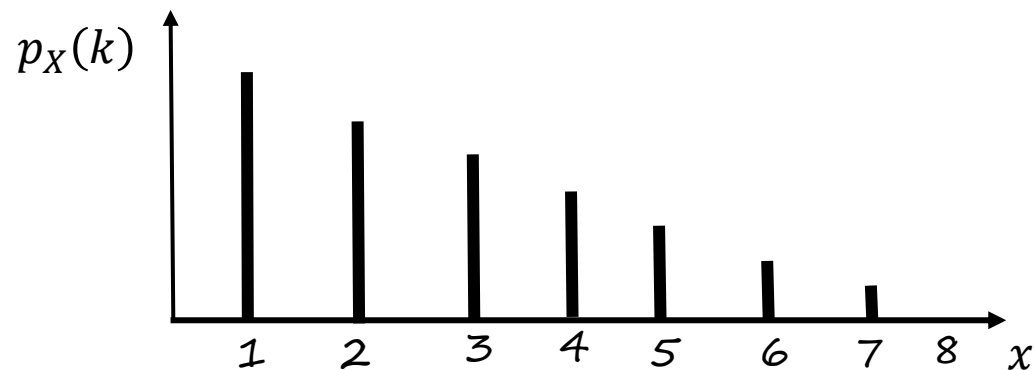
Total Expectation theorem*

*Also valid when Y is a discrete R.V. that ranges over an infinite set, as long as $E[|X|] < \infty$

Conditioning a geometric R.V.

- **X**: number independent tosses of a coin until first heads with $P(\text{Heads}) = p$

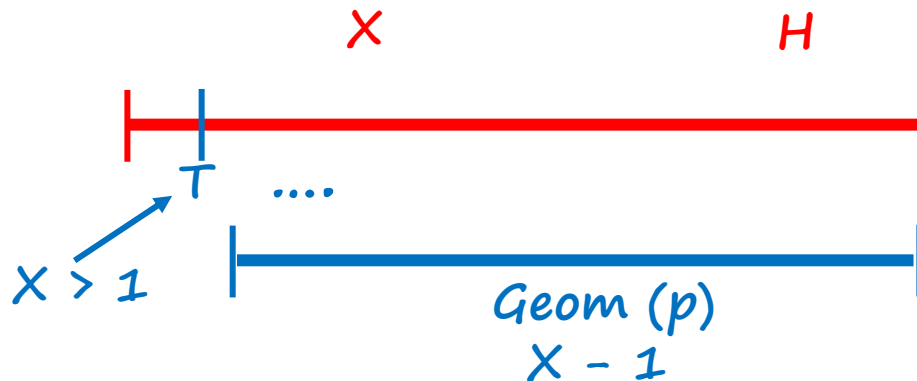
$$p_X(x) = (1 - p)^{x-1}p \quad x = 1, 2, 3, \dots$$



Memorylessness:

Number of **remaining** coin tosses, conditioned on Tails in the first toss, is **Geometric**, with parameter p

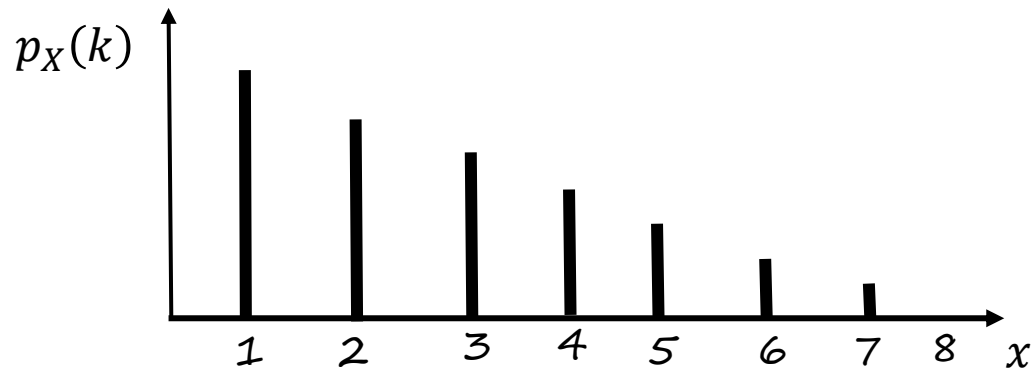
Conditioned on $X > 1$, $X - 1$ is geometric with parameter p



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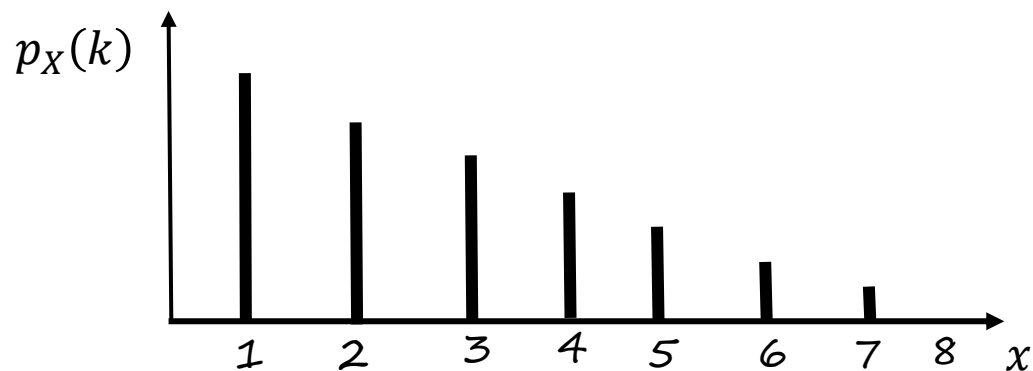
As an example

$$\begin{aligned} P_{X-1|X>1}(3) &= P(X - 1 = 3 \mid X > 1) = P(T_2 T_3 H_4 \mid T_1) = P(T_2 T_3 H_4) \\ P_{X-1|X>1}(k) &= P_X(k) &= (1-p)^2 \cdot p &= P_X(3) \end{aligned}$$

Conditioning a geometric R.V.

- **X**: number independent tosses of a coin until first heads with $P(\text{Heads}) = p$

$$p_X(x) = (1 - p)^{x-1}p \quad x = 1, 2, 3, \dots$$



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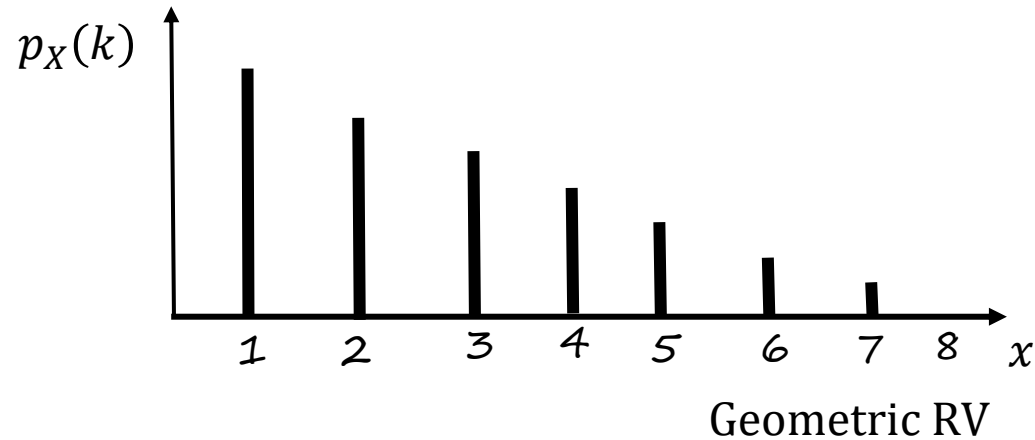
$$P_{X-1|X>1}(k) = P_X(k)$$



$$P_{X-n|X>n}(k) = P_X(k)$$

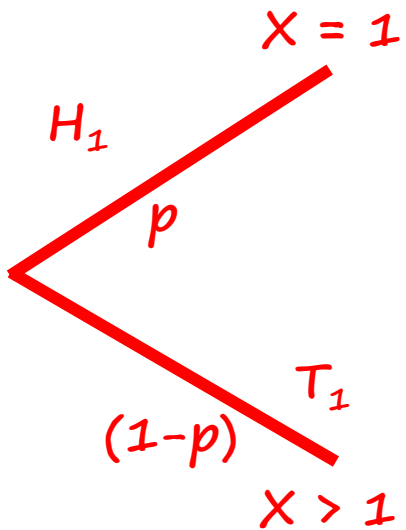
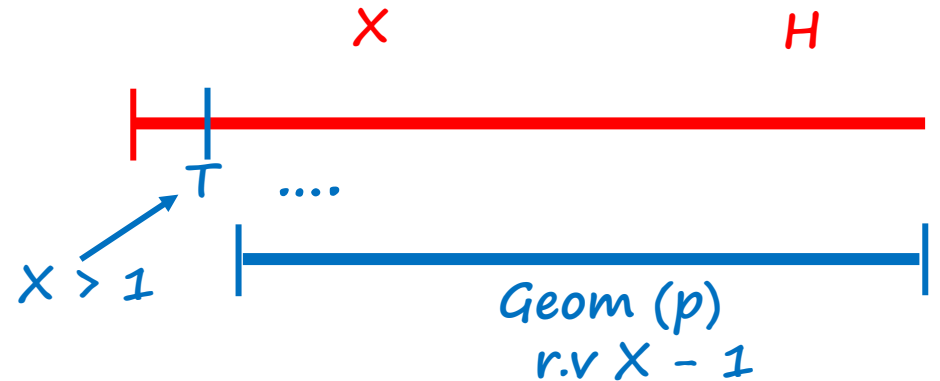
Conditioned on $X > n$, $X - n$ is geometric with parameter p

The mean of the geometric



$$E[X] = \sum_{k=1}^{\infty} k p_X(k) = \sum_{k=0}^n k (1-p)^{k-1} p$$

$$E[X] = \frac{1}{p}$$



$$\Rightarrow E[X] = 1 + E[X - 1]$$

$$\Rightarrow = 1 + p E[X - 1 \mid X = 1] + (1-p) E[X - 1 \mid X > 1]$$

$$\Rightarrow E[X] = 1 + 0 + (1-p) E[X]$$

$$\Rightarrow E[X] = 1 + 0 + E[X] - p \cdot E[X]$$

The mean of the binomial

- X : binomial with parameters n, p

❖ number of successes in n independent trials

$X_i = 1$ if i^{th} trial is a success; p
(indicator variable)
 $X_i = 0$ otherwise $1-p$

$$E[X] = np$$

$$X = X_1 + \dots + X_n$$

$$E[X] = \underbrace{E[X_1]}_p + \dots + \underbrace{E[X_n]}_p = np$$

$$E[X] = \sum_{k=0}^n k p_X(k) = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$$

Exercise No 02!



Example ($n = 5$)



$\{H_1, T_2, T_3, T_4, H_5\}$

Suppose you toss a biased coin n times with parameter ' p ' (for Head). Define the following random variables X , Y and Z :

X = The number of heads in n throws

$$Y = 3X + 4$$

$$Z = X^2$$

$X =$

$Y =$

$Z =$

- Calculate Expected value of X , Y and Z .

$$\blacksquare E[X] = np$$

$$\blacksquare E[Y] = 3np + 4$$

$$E[Z] = \sum_{k=0}^n k^2 \binom{n}{k} p^k (1-p)^{n-k}$$